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**Task Description:**

Pulmonary airway closure is a potentially dangerous event that can occur in microgravity environments and may result in limited gas exchange for flight crew during long-term space flight. To reopen collapsed airways, a semi-infinite bubble of air must separate airway walls that are held in apposition by a viscous lining fluid. However, the fluid mechanical stresses exerted by the migrating bubble can be very large and damage the lung. The fluid mechanics issues related to these studies are complex and are intimately related to the physiological responses of the lung. We will investigate the cellular responses to interfacial and viscous stresses that occur as a surfactant-doped bubble progresses over pulmonary epithelial cells that line the airways and alveoli in the lung. By identifying the magnitude of these stresses and the physiological response to the stimulus, we will provide insight of appropriate methods for opening collapsed airways, and how these methods depend upon the physicochemical properties of the airway lining fluid. As such, our studies will develop the link between fluid physics and potentially hazardous life-science issues that may be manifested during long-term space flight. Specific Aim #1: Study the fluid dynamical and surfactant physicochemical responses related to the opening of collapsed pulmonary airways in a microgravity environment.
Specific Aim #2: Examine cellular trauma due to bubble progression in a physical model of a collapsed airway.

Specific Aim #3: Predict flow and surfactant behavior that will minimize cellular injury during the opening of collapsed airways. Our studies are of importance to NASA for two reasons. First, pulmonary airway closure is more ubiquitous in a microgravity environment than on earth - this has been demonstrated in microgravity life-science studies of pulmonary function during short-term parabolic flights and long-term investigations on the SpaceLab (SLS-1). In microgravity, airway closure exists throughout the lung, rather than only in the basal 'gravitationally dependent'regions. While this closure occurs primarily at low lung volumes, recent parabolic flight studies of inhaled gas boluses indicate that airway closure in microgravity occurs in close proximity to airways that remain open, and may provide the inhomogeneity in the lung. Such closure may impact pulmonary function in microgravity environments during 'normal' and disease conditions. It is thus important to understand the complex fluid-structure interaction of this system to determine the likelihood of pulmonary cellular damage in microgravity,and to identify methods of reducing the effects of these stresses on pulmonary cells.

In addition, airway closure occurs in disease, and thus further understanding of the interaction between air-liquid interface behavior and pulmonary cellular responses could lead to advanced therapeutics for individuals on earth. The small physical size of the airway/bubble system that exists in regions of airway closure means that gravitational effects are small. While flight studies of this system are not proposed herein, we may in the future find it important to scale-up this system so as to measure the convection, transport and stress behavior in a system of larger physical size. Such studies would have to be performed in a microgravity environment to remove the effects of gravity. However, it is important to develop a basic understanding of these systems in a terrestrial environment so that we can determine the scaling principles from which the microgravity flight studies would be based.

**Rationale for HRP Directed Research:**

**Research Impact/Earth Benefits:**

Specific Aim #1: Study the fluid dynamical and surfactant physicochemical responses related to the opening of collapsed pulmonary airways in a microgravity environment.
• We have identified the magnitude and duration of mechanical stresses that exist on airway epithelial cells during airway reopening. These computational fluid mechanics calculations are incorporated into manuscripts that have been published in the Journal of Applied Physiology (Bilek et al. 2003 and Kay et al. 2004). These studies also investigate the responses of cells to these mechanical stresses, and thus are described in more complete detail in Specific Aim #2, below.

We have completed studies to predict the stress variation that occurs over cells due to topographical modifications. These computations have been published in Jacob, A.M. and D. P. Gaver III, An investigation of the influence of cell topography on epithelial mechanical stresses during pulmonary airway reopening. Physics of Fluids, 17: 031502, 2005 which was in a special Biofluids issue of the Physics of Fluids. The abstract is as follows: Jacob, A.M. and D. P. Gaver III, An investigation of the influence of cell topography on epithelial mechanical stresses during pulmonary airway reopening. In press, Physics of Fluids. The goal of this study is to assess the local mechanical environment of the pulmonary epithelium in a computational model of airway reopening. To this end, the boundary element method (BEM), in conjunction with lubrication theory, is implemented to assess the stationary-state behavior of a semi-infinite bubble traveling through a liquid-occluded parallel plate flow chamber lined with epithelial cells. The fluid occlusion is assumed to be Newtonian and inertia is neglected. The interactions between the microgeometry of the model airway's walls and the interfacial kinematics surrounding the bubble's tip result in a complex, spatially- and temporally-dependent stress distribution. The walls' non-planar topography magnifies the normal and shear stresses and stress gradients. We find that decreasing the bubble's speed serves to increase the maximum normal stress and stress gradient while decreasing the maximum shear stress and stress gradient. Our results give credence to the pressure-gradient-induced epithelial damage theory recently proposed by Bilek et al. [J. Appl. Physiol. 94, 770 (2003)] and Kay et al. [J. Appl. Physiol. (2004)]. We conclude that not only may the normal stress gradient likely be responsible for epithelial damage in this model of airway reopening, the stress magnitude may be even more severe than had been originally predicted.

• Physico-chemical hydrodynamic issues have been investigated using the a new technique that we developed (Dual-Reciprocity Boundary Integral Method), which accurately computes convection-diffusion interactions in free-surface boundary value problems such as those occurring during airway reopening. Our paper on this topic was published in the Journal of Fluid Mechanics (Ghadiali and Gaver, 2002), and the abstract is as follows: Ghadiali, S.N. and D. P. Gaver III, The influence of non-equilibrium surfactant dynamics on the flow of a semi-infinite bubble in a rigid cylindrical tube. J. Fluid Mech., 478: 165-196, 2003. We have performed a computational investigation of the progression of a semi-infinite air bubble in a surfactant doped, fluid-filled rigid capillary, intended to mimic the continual interfacial expansion dynamics that occur during the opening of collapsed pulmonary airways. This model is capable of simulating mixed-kinetic conditions with non-linear surfactant equations of state similar to those of pulmonary surfactant. Several dimensionless parameters govern the system responses. These are: the capillary number (Ca) that relates viscous to surface tension forces; the elasticity number (El), a measure of the ability of surfactant to modify the surface tension; the bulk Péclet number (Pe), relating bulk convection rates to diffusion rates; the adsorption and desorption Stanton numbers (Sta and Std) that relate the adsorption/desorption rates to surface convective rates, and finally the adsorption depth (8), a dimensionless bulk surfactant concentration parameter. We investigate this model by

**Task Progress:**

performing exhaustive parameter variation studies. We find that the surfactant properties can strongly influence the interfacial pressure drop through modification of the surface tension and the creation of Marangoni stresses that influence the viscous stresses along the interface. In addition, these studies demonstrate that, depending upon the range of parameters, either film thickening or film thinning responses are possible. In particular, we find that when Pe>>1 (as with pulmonary surfactant) or concentrations are low, bulk concentration profiles can substantially differ from near-equilibrium approximations and can result in film-thinning. These responses may influence stresses on epithelial cells that line pulmonary airways, the stability of these airways, and may be important to the delivery of surfactant to pulmonary airways and alveoli.

• We have completed a preliminary set of computations to investigate transport behavior during pulsatile reopening. These studies are described in Specific Aim #3, below.

Specific Aim #2: Examine cellular trauma due to bubble progression in a physical model of a collapsed airway.

We have been very successful in completing studies associated with this specific aim. To date, two papers have been published that identify the stimulus-response behavior of pulmonary epithelial cells that are exposed to a propagating air-liquid interface. The two papers and their abstracts that have been published in this regard are:

Bilek, A.M., K.C. Dee and D. P. Gaver, Mechanisms of surface-tension-induced epithelial cell damage in a model of pulmonary airway reopening. Journal of Applied Physiology, 94: 770-783, 2003. Airway collapse and reopening due to mechanical ventilation exerts mechanical stress on airway walls and injures surfactant-compromised lungs. The reopening of a collapsed airway was modeled experimentally and computationally by the progression of a semi-infinite bubble in a narrow fluid-occluded channel. The extent of injury caused by bubble progression to pulmonary epithelial cells lining the channel was evaluated. Counter-intuitively, cell damage increased with decreasing opening velocity. The presence of pulmonary surfactant, Infasurf, completely abated the injury. These results support the hypotheses that mechanical stresses associated with airway reopening injure pulmonary epithelial cells and that pulmonary surfactant protects the epithelium from this injury. Computational simulations identified the magnitudes of components of the stress cycle associated with airway reopening (shear stress, pressure, shear stress gradient, or pressure gradient) that may be injurious to the epithelial cells. By comparing these magnitudes to the observed damage, we conclude that the steep pressure gradient near the bubble front was the most likely cause of the observed cellular damage.

Kay, S.S, A.M. Bilek, K. C Dee and D. P. Gaver III, Pressure gradient, not exposure duration, determines the extent of epithelial cell damage in a model of pulmonary airway reopening. Journal of Applied Physiology, 97: 269-276, 2004. The reduction of tidal volume during mechanical ventilation has been shown to reduce mortality of patients with acute respiratory distress syndrome (ARDS), but epithelial cell injury can still result from mechanical stresses imposed by the opening of occluded airways. To study these stresses, a fluid filled parallel-plate flow chamber lined with epithelial cells was used as an idealized model of an occluded airway. Airway reopening was modeled by the progression of a semi-infinite bubble of air through the length of the channel, which cleared the fluid. In our prior study, the magnitude of the pressure gradient near the bubble tip was directly correlated to the epithelial cell layer damage (Bilek et al., 2003). However, in that study it was not possible to discriminate the stress magnitude from the stimulus duration because the bubble propagation velocity varied between experiments. In the present study, the stress magnitude is modified by varying the viscosity of the occlusion fluid while fixing the reopening velocity across experiments. This approach causes the stimulus duration to be inversely related to the magnitude of the pressure gradient. Nevertheless, cell damage remains directly correlated with the pressure gradient, not the duration of stress exposure. The present study thus provides additional evidence that the magnitude of the pressure gradient induces cellular damage in this model of airway reopening. We explore the mechanism for acute damage and also demonstrate that repeated reopening and closure is shown to damage the epithelial cell layer even under conditions that would not lead to extensive damage from a single reopening event.

The most significant finding of these studies is the clear identification of the magnitude of normal stress as determining the degree of epithelial cell damage. This provides a paradigm-shift from the conventional (but unsubstantiated) assumption that shear-stress would cause epithelial damage. Clearly, this result is significant because it identifies the components of stress that must be minimized to protect individuals from pulmonary trauma during airway reopening.

Specific Aim #3: Predict flow and surfactant behavior that will minimize cellular injury during the opening of collapsed airways.

We have made significant progress on this specific aim. In particular, we have completed a computational investigation of a pulsatile bubble motion in a channel, and calculated the transport of a surface-associated contaminant as a preliminary model of pulmonary surfactant. This paper was published in:

Zimmer IV, M.E., H.A.R. Williams and D. P. Gaver III, The transport of an inactive surface-associated contaminant during the pulsatile motion of a semi-infinite bubble in a channel. J. Fluid Mech., 537: 1-33, 2005: We investigate a theoretical model of the pulsatile motion of a contaminant-doped semi-infinite bubble in a rectangular channel. We examine the fluid mechanical behavior of the pulsatile bubble, and its influence on the transport of a surface-inactive contaminant (termed surfinactant). This investigation is used to develop a preliminary understanding of surfactant responses during unsteady pulmonary airway reopening. Reopening is modeled as the pulsatile motion of a semi-infinite gas bubble in a horizontal channel of width 2a filled with a Newtonian liquid of viscosity m and constant surface tension g. A modified Langmuir sorption model is assumed, which allows for the creation and respreading of a surface multilayer. The bubble is forced via a time-dependent volume flux Q(t) with mean and oscillatory components (QM and Q omega, respectively) at frequency omega. The flow behavior is governed by the dimensionless parameters: CaM = μ QM/(2a gamma), a steady-state capillary number, which represents the ratio of viscous to surface tension forces; CaOmega = μ Q omega/(2a gamma), an oscillatory forcing magnitude; Omega = omega μa/gamma, a dimensionless frequency that represents the ratio of viscous relaxation to oscillatory-forcing timescales; and A=2CaOmega/Omega, a dimensionless oscillation amplitude. Our simulations indicate that contaminant deposition and retention in the bubble cap region occurs at moderate frequencies if retrograde bubble motion develops during the oscillation cycle. However, if oscillations are too rapid the ensuing large forward tip velocities cause a net loss of contaminant from the bubble tip. Determination of an optimal oscillation range may be important in reducing ventilator-induced lung injury associated with infant and adult respiratory distress syndromes by increasing surfactant transport to regions of collapsed airways.

The research associated with this specific aim has identified the existence of pulsatile flow parameters that will

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| | encourage for formation of large concentrations of surfactant, which may reduce the magnitudes of stress on airway epithelial cells. Further computations will investigate the coupled behavior that exists with surface-activity – this aspect is one of the primary goals of the final year of the project. Furthermore, we are completing experiments using capillary tubes as models of airways to identify asymmetric waveforms that will reduce the pressures necessary to reopen pulmonary airways. |
| **Bibliography Type:** | Description: (Last Updated: 10/21/2009) |
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| **Abstracts for Journals and Proceedings** | Gaver DP. "Multi-scale effects of surfactant transport during pulsatile airway reopening. " Annual Meeting of the Biomedical Engineering Society, October 2005.<br>Abstracts of the Annual Meeting of the Biomedical Engineering Society, 2005. , Oct-2005 |
| **Abstracts for Journals and Proceedings** | Smith BJ, Gaver DP. "Airway reopening: oscillating air finger propagation through a liquid-filled tube." Annual Meeting of the Biomedical Engineering Society, October 2005.<br>Abstracts of the Annual Meeting of the Biomedical Engineering Society, 2005. , Oct-2005 |
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| **Articles in Peer-reviewed Journals** | Zimmer ME, Williams HA, Gaver DP 3rd. "The pulsatile motion of a semi-infinite bubble in a channel: flow fields, and transport of an inactive surface-associated contaminant." J Fluid Mech. 2005 Aug;537: 1-33. , Aug-2005 |
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